S&P 500 FACTOR INVESTING

A Multi-Strategy Quantitative Research Project

- Eight systematic strategies built on quality, value, momentum, and trend factors
- One-change-at-a-time experimental ladder — every improvement is scientifically readable
- Backtrader simulation with native Stop/Limit OCO orders · Zero commission baseline
- Full robustness battery: Monte Carlo, block bootstrap, walk-forward, Hansen SPA
- Data window: S&P 500 constituents, June 2006 – May 2026 · Eval window: May 2016 – May 2026
- Group participant: Ömer Faruk Yıldız

PAPER SUMMARY

Han, Zhou & Zhu (2016) — 'A Trend Factor' — Journal of Financial Economics

Core Idea

The Trend Factor:

Normalized moving averages across 11 horizons (3→1,000 trading days) capture short-term reversal, medium-term momentum, and long-term mean reversion in one signal.

Methodology:

Each month, cross-sectional OLS regression of stock returns on 11 MA ratios; coefficient vector smoothed with 12-month trailing average → stock-specific predicted return.

Result in Paper:

Top-minus-bottom long-short trend portfolio earns significant positive alpha; the factor is orthogonal to Fama-French factors.

Our Adaptation:

We apply the same concept to a long-only S&P 500 universe and combine the trend signal with quality (ROE), value (P/E), and momentum (12m price return).

Two Implementations:

trend_z (base) — full-sample index regression (look-ahead biased) · trend_expanding_z (imp 1–5, 7) — expanding window (honest) · trend_hzz_z (imp 6) — paper-faithful cross-sectional

Source: Han, Zhou & Zhu (2016); docs/PROJECT_REPORT.md §2

DATA & SIGNALS

Frozen CSVs — no live downloads during backtests

File	Source	Rows	Span	Description
sp500_prices_long.csv	Tiingo	2,293,664 rows	2006–2026	Daily OHLCV adj. prices, 503 tickers
sp500_fundamentals_daily_long.csv	Tiingo	2,295,967 rows	2006–2026	Point-in-time mkt cap, P/E, P/B
sp500_fundamentals_statements_lo	Tiingo	3,459,358 rows	2005–2026	Statement data keyed by date_available (no
sp500_index_yahoo.csv	Yahoo ^GSPC	5,030 rows	2006–2026	Frozen benchmark OHLCV
sp500_constituents.csv	Wikipedia	503 tickers	current	GICS sector, date added

Four Factors (all month-end, winsorized, z-scored, lagged 1 month):

- **ROE (Quality):** roe_z — latest filing from date_available join — no look-ahead
- **-P/E (Value):** pe_z — inverted so high z = cheap
- **Momentum:** momentum_z — 12-month total price return
- **Trend:** trend_z / trend_expanding_z / trend_hzz_z — see Paper slide

[!] Survivorship bias: universe = current S&P 500 — failed/removed names absent.

Source: data/raw/; project_core.py: build_factor_panel()

STRATEGY LADDER — 8 VARIANTS

One-change-at-a-time principle: each step modifies exactly one design dimension

#	Strategy Name	Foundation	Change	Verdict
0	base_equal_top10	—	Full-sample trend (look-ahead biased)	[doc] Assignment baseline
1	improved_1_expanding_trend	base	Trend → expanding window (no look-ahead)	[OK] Honest baseline
2	improved_2_stop_take	imp 1	Add 10% stop / 20% take-profit	[OK] Risk mgmt layer
3	improved_3_dynamic_ic	imp 2	Rolling rank-IC weights (50% shrinkage)	[X] Didn't help
4	improved_4_walkforward	imp 2	Walk-fwd stop/take → 5% / 30% (6×6 grid)	[*] Best risk-managed
5	improved_5_regime_filter	imp 4	10m SMA regime filter (cash when below)	[X] Hurt performance
6	improved_6_hzz_trend	imp 4	Trend → HZZ cross-sectional (paper-faithful)	[OK] Paper faithful
7	improved_7_costs	imp 4+6	Time-varying cost sensitivity (not a strategy)	[chart] Cost study
8	improved_8_equal_weight_top20	imp 4	Top-N → 20 + sizing → 5% of equity (1/N)	[wealth] Wealth-maximizer

[*] = project finalist on risk-adjusted return · [X] = documented failed experiment (retained for transparency)

EXECUTION MODEL & CODE ARCHITECTURE

Backtrader physical broker simulation — two engines, three position sizers

Initial capital:	$1,000,000
Rebalance frequency:	Monthly (signal at month-end → fill at next open)
Entry / exit:	bt.Order.Market entries; bt.Order.Stop + bt.Order.Limit (OCO) exits
Commission (baseline):	0 bps (Improved 7 adds time-varying costs)
Long-only assertion:	Checked after every Backtrader run; aborts on any short position
Sizing — imp 1-7:	FixedCashSizer: $100,000 per position
Sizing — imp 8:	EquityPercentSizer: 5% of current equity per position (1/N for top-20)
Stop/Take — imp 2-3:	10% stop / 20% take (round numbers, not optimised)
Stop/Take — imp 4-8:	5% stop / 30% take (walk-forward selected from 6×6 grid)

Improved 8 spec — one config object:

```
# StrategySpec — one dataclass per variant
core.StrategySpec(
    name = core.IMPROVED_8_STRATEGY_NAME,
    weights = {'roe':1,'pe':1,'momentum':1,'trend':1},
    top_n = 20,  trend_col = 'trend_expanding_z',
    stop_loss = 0.05,  take_profit = 0.30,
    sizing_method = 'percent_of_equity',
    sizing_target_pct = 0.05
)
```

All 8 variants = StrategySpec + runner.
Declarative, testable, reproducible.

Source: project_core.py: StrategySpec, run_backtrader_daily_stop_take()

KEY METHODOLOGY — COMMON EVALUATION WINDOW

The most important methodological fix in this project

Problem:

Different strategies have different warmup periods:
 · base uses full-sample trend_z → trades from 2010-05
 · imp 1–5, 7 use trend_expanding_z → trades from 2016-05 (72-month index regression warmup)
 · imp 6 uses trend_hzz_z → trades from 2011-05

Original code computed Sharpe over all 240 months. Pre-warmup months = zero returns.
This dilutes the mean more than the std, giving longer-warmup strategies artificially low Sharpes.

Fix:

EVALUATION_START = 2016-05-31 (121 months through 2026-05-31)
All Sharpes, drawdowns, Monte Carlo, bootstrap, walk-forward, and benchmark comparison restricted to this common window across all 8 strategies.

Academically standard:
Every published factor paper
restricts cross-strategy comparison
to a common evaluation window
(Fama-French, AQR, HZZ paper itself).

Strategy	Old Sharpe (240m)	Fixed Sharpe (121m)	Change
base	1.13	1.20	+0.07
improved_1	0.82	1.18	+0.36 ← biggest gain
improved_4	0.80	1.15	+0.35
improved_6	0.81	1.02	+0.21

Source: project_core.py: metrics_over_evaluation_window(), filter_to_evaluation_window()

FACTOR ANALYSIS — WHICH SIGNALS WORK?

Factor-mimicking portfolios: each factor in isolation (long-short, eval window)

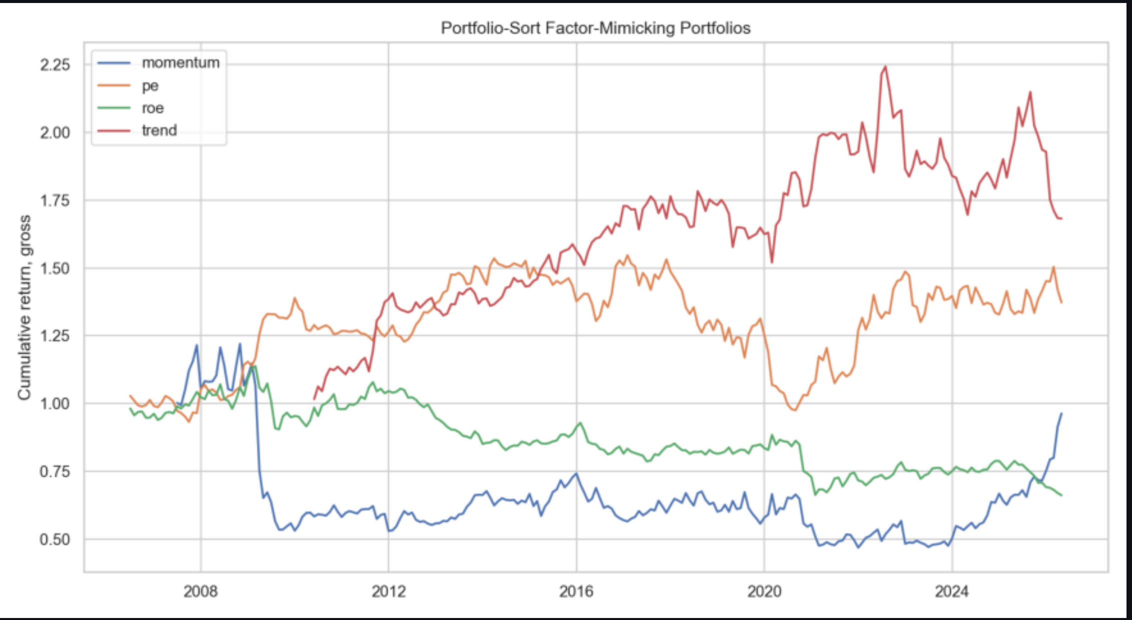
Factor	Ann. Sharpe	t-stat	Avg Rank IC	Role in strategy
Trend (cross-section HZZ)	0.39	1.56	0.022	Strongest — dominates the composite
Trend (index expanding)	0.36	1.44	0.022	Strong — used in imp 1–5, 7
Value (−P/E)	0.21	0.92	0.012	Weak alone; reduces variance in combo
Momentum (12m)	0.08	0.34	0.004	Near-zero standalone; combo diversifier
Quality (ROE)	−0.21	−0.92	0.004	Negative alone on this sample!

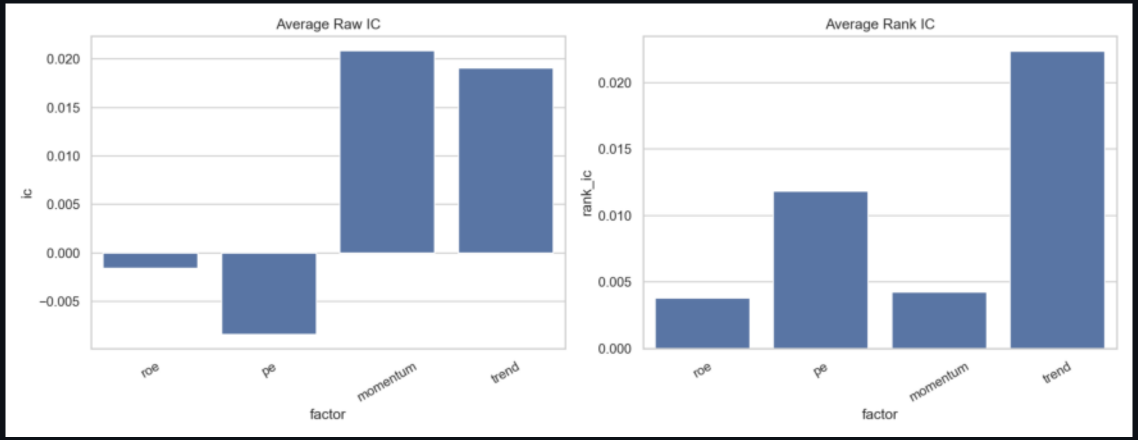
- Trend is the engine — the two-factor combo of trend+momentum alone explains most of the composite Sharpe.
- Improved 3 tried dynamic rank-IC weighting to let the data pick factor weights — result: no improvement (too little signal history).
- Negative ROE standalone Sharpe means quality is contrarian on this S&P 500 sample; it mainly reduces left-tail volatility.
- Composite Sharpe (>1.0) materially exceeds best single-factor Sharpe (0.39) — diversification benefit is real.

Source: results/fmp_analysis/; project_core.py: make_fmp_returns(), summarize_fmps()

FACTOR PERFORMANCE — CHARTS

Equity curves and IC summaries





Left: cumulative returns of factor-mimicking portfolios (long-short, each factor in isolation).

Right: rank-IC distribution across months — trend has the most consistently positive IC.

RESULTS — HEADLINE COMPARISON

All 8 strategies · Eval window 2016-05 to 2026-05 · 121 months

Strategy	Vec Sharpe	Ann Return	Max DD	Final $	Avg Pos
Base (full-sample, look-ahea	1.20	7.2%	-7.3%	$2.03M	9.9
Imp 1 (expanding trend)	1.18	13.1%	-15.8%	$3.51M	8.4
Imp 2 (+stops 10/20%)	1.05	10.5%	-13.0%	$2.72M	8.4
Imp 3 (+dyn IC weights) -	0.97	10.2%	-14.1%	$2.64M	8.4
Imp 4 (walk-fwd 5/30%) [sel	1.15	10.5%	-7.4%	$2.75M	8.4
Imp 5 (regime filter) -	0.93	8.0%	-11.2%	$2.15M	7.0
Imp 6 (HZZ trend)	1.02	7.3%	-8.0%	$2.04M	9.9
Imp 8 (eq-wt top-20)	1.03	13.7%	-10.9%	$3.63M	16.9

[sel] Improved 4: best Sharpe (1.15) and shallowest drawdown (−7.4%) — walk-forward selected 5%/30% stops.

[wealth] Improved 8: highest final equity ($3.63M) via equal-weight compounding across ~17 names.

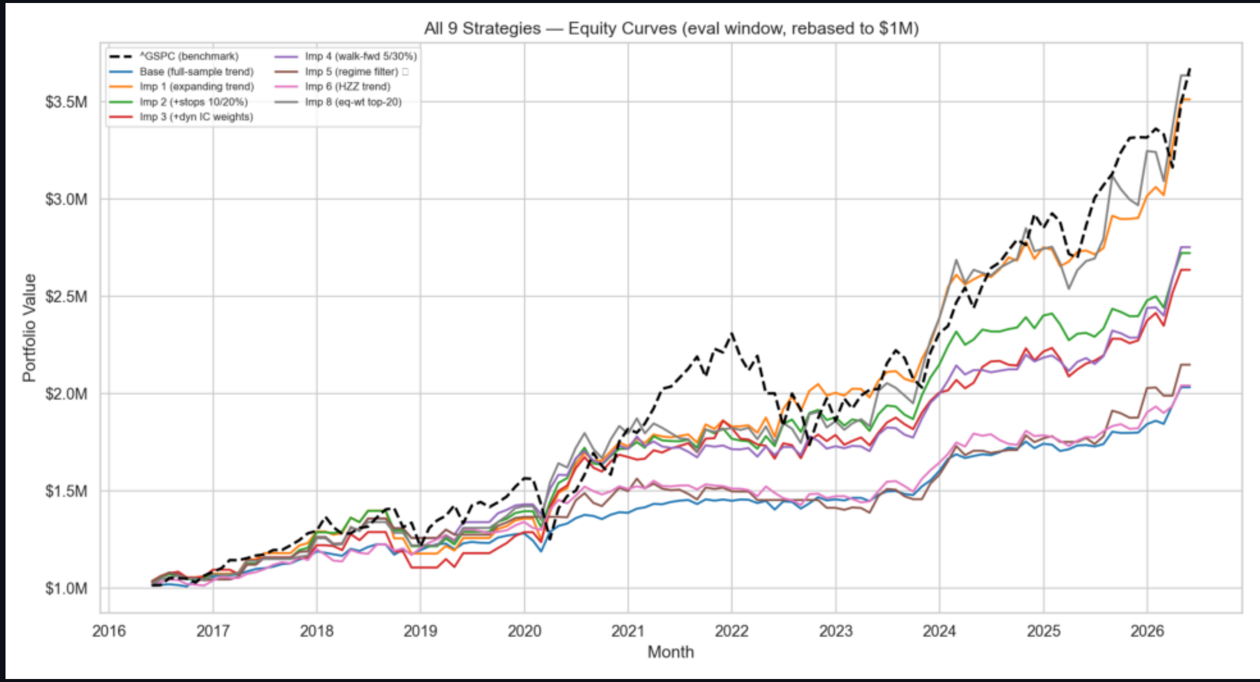
[!] Base greyed — full-sample trend is look-ahead biased; Sharpe not directly comparable to others.

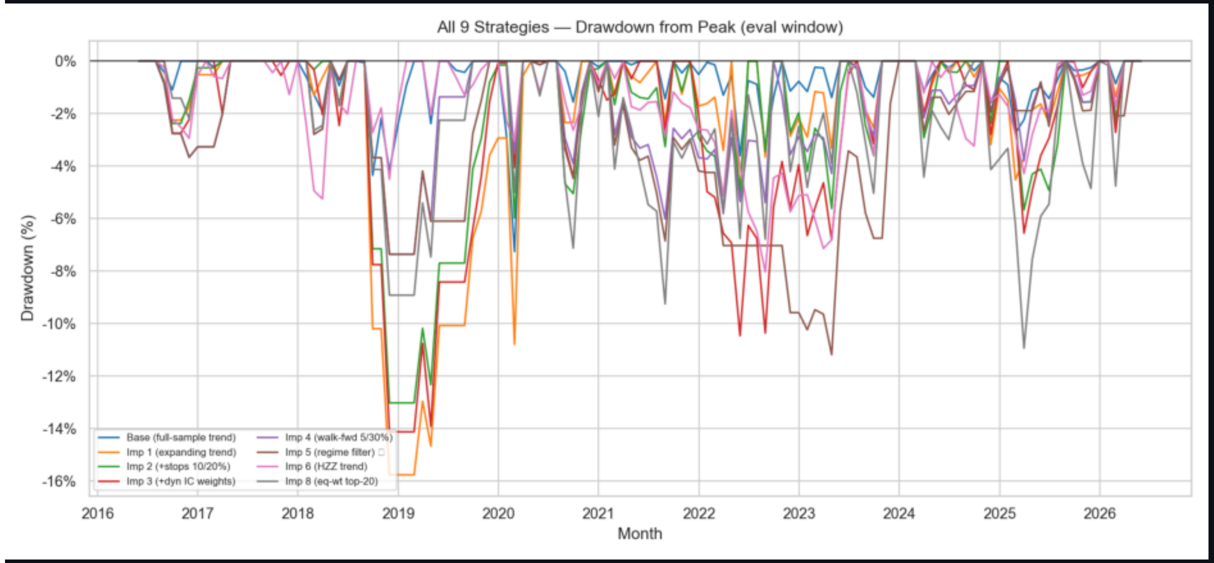
Note: vector final equity restated from $1M at eval start. Sharpes use eval window in all engines.

Source: results/comparison/all_strategies_metrics.csv

RESULTS — EQUITY & DRAWDOWN CURVES

All 8 strategies · eval window



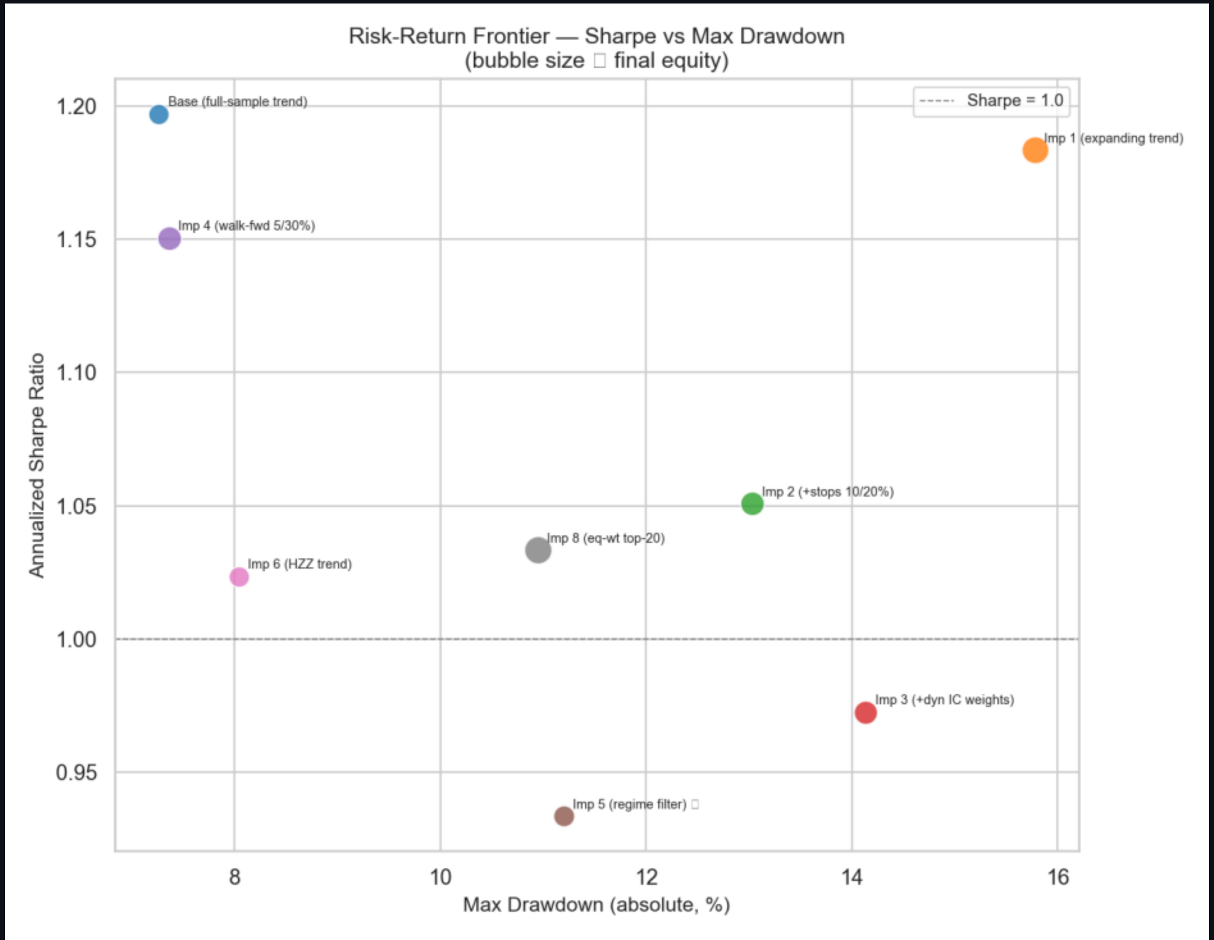


Left: equity curves (vector, rebased $1M at eval start). All strategies positive.

Right: drawdown from peak. Imp 4 shallowest; Imp 8 deepest (more capital deployed).

RESULTS — RISK-RETURN FRONTIER

Sharpe vs Max Drawdown — bubble size ∝ final equity



Key Reading:

Top-right corner: = best risk-adjusted return

Imp 4: Highest Sharpe + shallowest DD → efficier

Imp 8: Moves down-left deliberately: trades Sharp

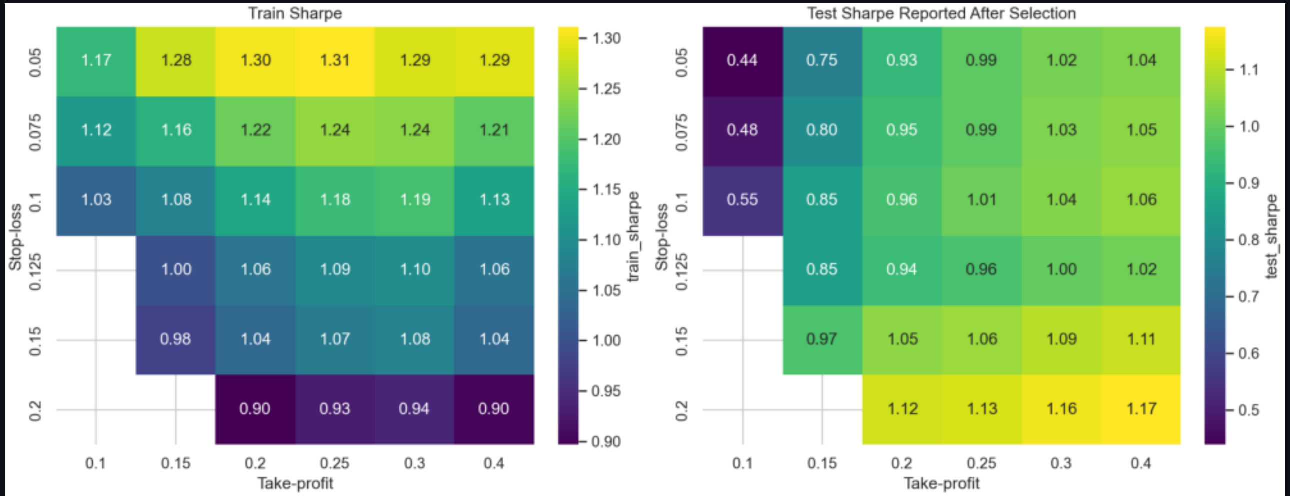
Base: Top-right only because of look-ahead bias

Imp 5, 3: Regime filter and dynamic IC — both hurt;

Source: figures/sharpe_vs_drawdown_scatter.png; results/comparison/all_strategies_metrics.csv

WHY IMPROVED 4 — WALK-FORWARD STOP/TAKE SELECTION

Data-driven risk-exit calibration on training data only



Methodology:

6×6 grid of (stop%, take%) combinations

Evaluated on training data ≤ 2020-12 only

Selection criterion: highest train-period Sharpe

Selected: 5% stop, 30% take-profit

Out-of-sample test Sharpe: 1.02 (honest check)

Result: drawdown drops to −7.4%; Sharpe rises to 1.15

Important: the grid search itself consumes

some of our 'multiple testing budget' — disclosed

transparently in the Limitations section.

Source: results/improved_strategy_4/; figures/improved4_stop_take_sensitivity_heatmap.png

ROBUSTNESS — VALIDATION BATTERY

Six independent lenses — each interrogates a different failure mode

1. Monte Carlo	1,000 random portfolios / strategy from same eligible universe. p = fraction beating strategy Sharpe. → base (0.013), imp_1 (0.019) clear $p<0.05$; imp_4 (0.088) borderline.
2. Block Bootstrap	1,000 paths resampling 6-month blocks of the strategy's own returns. → Characterises Sharpe and DD distribution without historical ordering.
3. Walk-Forward	Train ≤ 2020-12, test 2021+. Select on train; report test. → Imp_4 selected on train (1.29); imp_1 has highest test Sharpe (1.34).
4. Benchmark Alpha	OLS regression of excess return on ^GSPC (eval window). → All 8 strategies: statistically significant alpha at 1% level. Negative beta.
5. Cost Sensitivity	Time-varying commission + slippage 2006–2026. Three scenarios. → Both imp_4 and imp_6 survive pessimistic costs; Sharpe drop only 0.03–0.07.
6. Hansen SPA + Romano-Wolf	Multi-comparison correction across all 8 variants. 10,000 bootstrap reps. → [OK] IMPLEMENTED — SPA consistent p = 0.674 (cannot reject no-outperformance after correction).

Source: results/robustness/; results/comparison/all_strategies_*.csv

ROBUSTNESS — ALPHA & WALK-FORWARD

All 8 strategies covered (not just base + improved 1–3)

Benchmark Alpha vs ^GSPC (eval window, all 8 strategies):

Strategy	Ann Alpha	t-stat	p-value	Beta	Sig.
Base (full-sample, lo	7.91%	4.43	0.0000	-0.048	+ sig
Imp 1 (expanding tr	14.16%	3.96	0.0001	-0.073	+ sig
Imp 2 (+stops 10/20	11.52%	3.65	0.0003	-0.075	+ sig
Imp 3 (+dyn IC weig	11.16%	3.11	0.0019	-0.069	+ sig
Imp 4 (walk-fwd 5/3	11.63%	4.12	0.0000	-0.081	+ sig
Imp 5 (regime filter)	7.47%	3.32	0.0009	0.034	+ sig
Imp 6 (HZZ trend)	7.86%	3.76	0.0002	-0.037	+ sig
Imp 8 (eq-wt top-20	15.41%	3.98	0.0001	-0.120	+ sig

Walk-Forward (train ≤ 2020-12, test 2021+):

Strategy	Train Sharpe	Test Sharpe	Test Cum Ret
Imp 4 (walk-fwd 5/3	1.29	1.02	60%
Imp 6 (HZZ trend)	1.20	0.85	35%
Imp 2 (+stops 10/20	1.14	0.96	59%
Imp 8 (eq-wt top-20	1.12	0.98	102%
Imp 5 (regime filter)	1.11	0.79	43%
Base (full-sample, lo	1.10	1.30	46%
Imp 1 (expanding tr	1.03	1.34	103%
Imp 3 (+dyn IC weig	0.98	0.98	57%

All 8 strategies produce statistically significant alpha vs ^GSPC at the 1% level. Negative beta = active factor tilt.

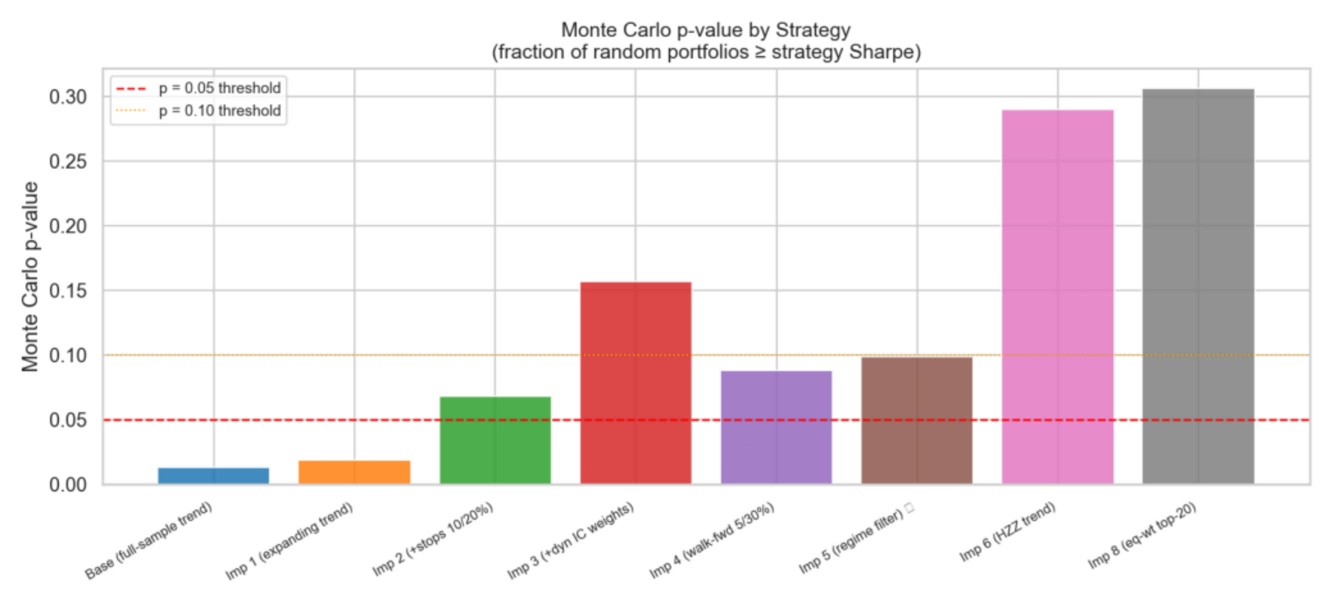
Walk-forward: Imp 4 selected on train (1.29); Imp 1 has highest out-of-sample test Sharpe (1.34). Single splits are noisy.

[!] Excess return p-values are unadjusted. Multi-comparison correction (Hansen SPA) on next slide.

Source: results/comparison/all_strategies_benchmark.csv, all_strategies_walk_forward.csv

ROBUSTNESS — MONTE CARLO & MULTI-COMPARISON CORRE

The honest truth about statistical significance



Testing 8 variants at $p<0.05$:

P(at least one false positive) = $1 - (1-0.05)^8 \approx 34\%$
Raw p-values are biased downward by the search-space size.
Only base and Imp 1 clear 0.05 individually — and even those are not significant after FWER correction.

Raw MC p-values:

```
Base (full-sample, loo  p = 0.013
Imp 1 (expanding trend  p = 0.019
Imp 2 (+stops 10/20%)   p = 0.068
Imp 3 (+dyn IC weights  p = 0.157
Imp 4 (walk-fwd 5/30%)  p = 0.088
Imp 5 (regime filter)   p = 0.099
Imp 6 (HZZ trend)       p = 0.290
Imp 8 (eq-wt top-20)    p = 0.306
```

Hansen (2005) SPA Test — ALL 8 STRATEGIES:

Consistent p-value = 0.6737 → CANNOT reject null

Romano-Wolf StepM: no strategy survives at 5% FWER

Source: results/robustness/hansen_spa_results.csv; src/run_multi_comparison_test.py

ROBUSTNESS — TRANSACTION-COST SENSITIVITY (IMPROVED

Time-varying commission + slippage, 2006–2026 · Three scenarios

Cost Schedule (central):

```
2006:  7.0 bps comm  /  6.0 bps slip  →  26 bps RT
2008:  6.0 bps comm  /  8.0 bps slip  →  28 bps (crisis)
2014:  2.5 bps comm  /  2.0 bps slip  →   9 bps RT
2020:  0.7 bps comm  /  2.5 bps slip  →   6.4 (Covid)
2026:  0.5 bps comm  /  1.0 bps slip  →   3.0 bps RT
```

Results:

Scenario	Imp 4 Sharpe	Imp 6 Sharpe
Zero cost	1.150	1.023
Central	1.118	0.985
Pessimistic	1.086	0.948

Both finalists survive pessimistic costs. Small Sharpe drop (~0.03–0.07) because monthly turnover is modest and trade size is tiny vs S&P 500 average daily volume.

Source: results/improved_strategy_7/; docs/SIZING_AND_MARGIN.md

WHAT WE LEARNED

Interpretation of results

Improved 4 — the Sharpe story

Best risk-managed variant. Sharpe 1.15 (vector) / 1.40 (Backtrader). Max DD −7.4%.
Walk-forward selected 5%/30% stops on training data through 2020. Concentrated (8–9 names).

Improved 8 — the wealth story

Highest vector final equity ($3.63M). Equal-weight 1/N across 20 names compounds dynamically.
Pays for it with lower Sharpe (1.03) and deeper drawdown (−10.9%). Industry analog: RSP ETF.

Trend is the dominant factor

FMP Sharpe ~0.36–0.39; strongest rank IC (0.022). Quality and value mainly reduce variance.
The composite materially beats any single factor (diversification benefit is real).

The common-window fix overturned a key finding

'Improved 6 is best' was a dilution artifact. After the fix, Improved 4 clearly leads
on risk-adjusted return. This is a lesson in methodology.

The honest bottom line (Hansen SPA)

After correcting for 8 variants × multiple testing, SPA $p = 0.674$.
Cannot reject 'no strategy outperforms the benchmark' at 5% FWER.
Strong individual-test results are consistent with data-mining luck.

THE DECISION — WOULD WE INVEST?

Honest investor framing

Real capital today? **No — not yet.**

Research watchlist / paper portfolio? **Yes — Imp 4 and Imp 8.**

Reasons to keep researching:

+ Economically sensible process (quality, value, momentum, trend)
+ Historically attractive Sharpes (1.0–1.4) on a 10-year window
+ Survives realistic transaction costs (Improved 7)
+ Negative beta — not just riding the market
+ Transparent, multi-lens validation battery
+ Walk-forward selected risk exits (not full-sample overfit)

Reasons not to allocate yet:

- Survivorship bias — biggest gap; current members only
- Multiple testing — SPA p = 0.674 (cannot reject data mining)
- Walk-forward train = 4.5 yrs; academic standard is 10+
- No sector / liquidity / beta concentration controls
- Fixed-cash sizing gives only 4–9 realized names (imp 1–7)
- No live broker integration or production-grade CI

Verdict: A cautious investor would not allocate real capital on this deck alone. Improved 4 is the most credible risk-managed candidate; Improved 8 the wealth-compounding variant. Neither is deployable until survivorship-free data, SPA-corrected significance, and integrated costs are added.

HONEST LIMITATIONS & FUTURE WORK

What we would do with more time

Survivorship bias:

Universe = current S&P 500 members. Delisted/failed companies absent.
Fix: WRDS/CRSP point-in-time membership + delisted prices. User has WRDS access.

Multiple testing corrected:

Hansen SPA p = 0.674 — cannot reject no-outperformance.
Unadjusted individual p-values are misleading without this correction. [OK] DONE

Walk-forward window:

4.5 years train / 5.5 years test. Academic standard: 10+ years each.
Constrained by signal warmup (trend_expanding_z needs 72 months).

Fixed-cash sizing gaps:

FixedCashSizer → Margin rejections; only 4–9 of 10 target names filled.
Improved 8 solves this via EquityPercentSizer.

Cost integration:

Improved 7 is a sensitivity layer, not integrated into every Backtrader run.
Next step: integrate time-varying commission directly into the default spec.

Future work (priority order):

1. Survivorship-free panel via WRDS/CRSP
2. Long-short paper-faithful HZZ Q5−Q1 portfolio
4. Sector-concentration caps
5. Top-N sensitivity grid under equal-weight sizing

Source: README §12, §16; docs/MULTI_COMPARISON_TEST.md